10.8 Semisimple Operators and the Jordan–Chevalley Decomposition

The Jordan canonical form says that an operator whose characteristic polynomial splits is "a diagonal matrix plus a nilpotent one" — but only after a basis has been chosen. This section turns that sentence into a statement about the operator itself: $T = S+N$ with S diagonalizable, N nilpotent and, crucially, $SN = NS$; and there is exactly one such pair. Before that we need a name for the operators that deserve to be called diagonalizable even when the field is too small to see it.

10.8.1 Diagonalizable once the field is big enough

Two operators fail to be diagonalizable for completely different reasons. The rotation $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$ has $p_{\mathbf{R}} = m_{\mathbf{R}} = x^2 + 1$, with no real root, so it has no eigenvectors at all (Example (A Real Matrix with Non-Real Eigenvalues)). Yet over $\mathbb{C}$ it becomes $\operatorname{diag}(i, -i)$: the operator was fine, the field was too small. The shear $\mathbf{J}_2(0)$ has $m = x^2$ and is not diagonalizable over **any** field, however large: its single eigenvalue 0 has $a = 2$ and $g = 1$, and enlarging the field changes neither.

What separates the two cases is visible in the minimal polynomial. For $\mathbf{R}$ it is x^2+1, irreducible over $\mathbb{R}$ and appearing to the first power. For $\mathbf{J}_2(0)$ it is x^2, the **square** of the irreducible x. Chapter 9 proved that T is diagonalizable exactly when m_T is a product of **distinct linear** factors (Theorem 9.8.2). Deleting the word "linear" from that criterion gives the notion we want.

A semisimple operator is one whose minimal polynomial is squarefree: no irreducible factor is repeated.

Definition 10.8.1. Semisimple Operator

Let V be a finite-dimensional vector space over F with $\dim V \geq 1$, and let $T \in \mathcal{L}(V)$. We say T is **semisimple** if

$$m_T = p_1 p_2 \cdots p_k$$

for **distinct** monic irreducible $p_1, \ldots, p_k \in F[x]$; equivalently, if **no** irreducible polynomial $p \in F[x]$ satisfies $p^2 \mid m_T$. A matrix $\mathbf{A} \in M_n(F)$ is **semisimple over** F if $T_{\mathbf{A}}$ is.

In words: factor m_T into irreducibles over F (Theorem 6.3.5). Semisimplicity says every exponent in that factorization is 1. The two descriptions agree: the exponents are all 1 exactly when no p^2 divides m_T, by uniqueness of factorization. Note the dependence on the field, which is built into the word "irreducible over F".

Examples.

- **Diagonalizable operators.** If T is diagonalizable, m_T is a product of distinct linear factors (Theorem 9.8.2), and linear polynomials are irreducible (Definition 6.3.1). So T is semisimple.
- **The rotation.** $\mathbf{R} \in M_2(\mathbb{R})$ above has $m_{\mathbf{R}} = x^2 + 1$, irreducible over $\mathbb{R}$ since it has no real root (Theorem 6.3.1). One factor, exponent 1: semisimple over $\mathbb{R}$, though not diagonalizable over $\mathbb{R}$.

- **Any operator with irreducible minimal polynomial**, for the same reason. In particular the companion matrix $\mathbf{C}(p)$ of an irreducible p is semisimple, since $m_{\mathbf{C}(p)} = p$ (Theorem 10.5.3).
- **The degenerate case.** On a space of dimension 1, $T = \lambda \operatorname{id}_V$ has $m_T = x - \lambda$: semisimple. So is id_V on any V, with $m = x - 1$.

Nonexample by minimal change. Over $\mathbb{R}$, replace $m = x^2 + 1$ by $m = (x^2 + 1)^2$: the companion matrix $\mathbf{C}((x^2+1)^2) \in M_4(\mathbb{R})$ has $m = (x^2+1)^2$ (Theorem 10.5.3), so the irreducible $x^2 + 1$ is repeated and the matrix is **not** semisimple. The clause that fails is "distinct": there is one irreducible factor, but it occurs twice.

Why this definition. The word *simple* names the case in which V has no T-invariant subspace except $\{0\}$ and V. That happens exactly when m_T is irreducible **and** $\dim V = \deg m_T$, for then every nonzero v has T-annihilator m_T and so $Z(v;T) = V$ (Theorem 10.5.2). *Semisimple* then means "a direct sum of simple pieces", which is what Theorem 9.8.1 followed by a cyclic decomposition (Theorem 10.6.2) delivers. The definition is the right one because it is exactly diagonalizability after the field has been enlarged enough to split m_T — provided the field is not pathological. Making that precise needs one small fact: the minimal polynomial does not change when the field grows.

Lemma 10.8.1. The Minimal Polynomial Does Not Notice a Larger Field

Let $K \subseteq L$ be fields and $\mathbf{A} \in M_n(K)$, $n \geq 1$. Then the minimal polynomial of $\mathbf{A}$ computed in $K[x]$ equals the minimal polynomial of $\mathbf{A}$ computed in $L[x]$.

Idea. The degree of $m_{\mathbf{A}}$ is the position of the first linear dependence in the list $\mathbf{I}, \mathbf{A}, \mathbf{A}^2, \ldots$ (Proposition 9.5.4). Whether a list of matrices with entries in K is dependent is decided by row reducing one matrix with entries in K, and row reduction does not care that a bigger field is available.

Proof. Write m^K and m^L for the two minimal polynomials, and $d = \deg m^K$, $e = \deg m^L$.

Since $m^K \in K[x] \subseteq L[x]$ annihilates $\mathbf{A}$, Theorem 9.5.1 gives $m^L \mid m^K$, so $e \leq d$.

For the reverse inequality, list the entries of $\mathbf{I}, \mathbf{A}, \ldots, \mathbf{A}^e$ as the columns of a matrix $\mathbf{B} \in M_{n^2 \times (e+1)}(K)$, one column per power. A linear combination $\sum_{i=0}^{e} c_i \mathbf{A}^i$ is the zero matrix exactly when $\mathbf{Bc} = \mathbf{0}$ for $\mathbf{c} = (c_0, \ldots, c_e)$. By Proposition 9.5.4 applied over L, the list $(\mathbf{I}, \mathbf{A}, \ldots, \mathbf{A}^e)$ is dependent over L, so $\mathbf{Bc} = \mathbf{0}$ has a nontrivial solution in L^{e+1}, and hence $\operatorname{rank}_L \mathbf{B} < e + 1$ (Theorem 3.5.7). Now reduce $\mathbf{B}$ to reduced row echelon form by row operations performed inside K (Theorem 3.2.3); the resulting matrix $\mathbf{R}$ has entries in K, and the definition of reduced row echelon form (Definition 3.2.4) makes no reference to the field, so $\mathbf{R}$ is also **the** reduced row echelon form of $\mathbf{B}$ over L, by the uniqueness in Theorem 3.3.4. The rank is the number of pivots of $\mathbf{R}$ (Corollary 3.5.6), a number read off $\mathbf{R}$ alone. Hence

$$\operatorname{rank}_K \mathbf{B} = \operatorname{rank}_L \mathbf{B} < e + 1,$$

so $\mathbf{Bc} = \mathbf{0}$ has a nontrivial solution in K^{e+1} as well. That solution gives a nonzero $f \in K[x]$ of degree at most e with $f(\mathbf{A}) = \mathbf{O}$, whence $d = \deg m^K \leq e$ by Theorem 9.5.1 and Proposition 6.1.2.

So $d = e$, and $m^L \mid m^K$ with both monic of the same degree forces $m^L = m^K$.

■

Proposition 10.8.2. Semisimple Means Diagonalizable After Enlarging the Field

Let K be a field of characteristic 0, let $\mathbf{A} \in M_n(K)$ with $n \geq 1$, and let $L \supseteq K$ be a field over which $m_{\mathbf{A}}$ splits. Then

$$\mathbf{A} \text{ is semisimple over } K \iff \mathbf{A} \text{ is diagonalizable over } L.$$

Proof. By Lemma 10.8.1 the minimal polynomial is the same polynomial $m = m_{\mathbf{A}}$ whether computed over K or over L. By Theorem 9.8.2, $\mathbf{A}$ is diagonalizable over L if and only if m is a product of distinct linear factors in $L[x]$, that is, if and only if m splits over L — which it does by hypothesis — with **no repeated root**.

($\Leftarrow$) Suppose m has no repeated root in L. If some irreducible $p \in K[x]$ satisfied $m = p^2 h$, then $p \mid m$ and m splits over L, so p has a root $\lambda \in L$; by Theorem 6.4.2, $\operatorname{mult}_\lambda(m) \geq 2 \operatorname{mult}_\lambda(p) \geq 2$, and λ would be a repeated root of m. So no such p exists and $\mathbf{A}$ is semisimple over K.

($\Rightarrow$) Suppose $m = p_1 \cdots p_k$ with $p_1, \ldots, p_k \in K[x]$ distinct monic irreducibles. The formal derivative is computed from the coefficients (Definition 6.4.3), so each p_i' is the same polynomial whether read in $K[x]$ or in $L[x]$.

No p_i has a repeated root in L. Since K has characteristic 0 and $\deg p_i \geq 1$, the derivative p_i' is nonzero of degree $\deg p_i - 1$, so $p_i \nmid p_i'$; as p_i is irreducible, Lemma 6.3.2 gives $\gcd(p_i, p_i') = 1$, and by Corollary 6.2.4 there are $a, b \in K[x]$ with $ap_i + bp_i' = 1$. If $\lambda \in L$ were a repeated root of p_i, then $p_i(\lambda) = p_i'(\lambda) = 0$ by Theorem 6.4.5 (a), applied in $L[x]$; evaluating the identity at λ would give $1 = 0$.

No two p_i share a root in L. For $i \neq j$, a monic divisor of the irreducible p_j is 1 or p_j, so $p_i \nmid p_j$, and Lemma 6.3.2 gives $\gcd(p_i, p_j) = 1$. Again by Corollary 6.2.4, $ap_i + bp_j = 1$ for some $a, b \in K[x]$, and a common root in L would give $1 = 0$.

Therefore, by Theorem 6.4.2, every root of m in L has multiplicity 1: it is a root of exactly one p_i, and a simple one. So m is a product of distinct linear factors over L, and $\mathbf{A}$ is diagonalizable over L.

■

Remark. A field over which every irreducible polynomial has distinct roots in a splitting field is called **perfect**. The proof above shows that every field of characteristic 0 is perfect, and every finite field is perfect as well, although that needs an argument we do not give here. Over a nonperfect field Proposition 10.8.2 can fail: in characteristic p there are irreducible polynomials of the form $q(x^p)$, whose derivative vanishes identically and which acquire repeated roots in a splitting field. Everything in the rest of this section takes place over a field where p_T splits, so this subtlety will not arise again.

Quick check. Over $\mathbb{Q}$, which of $\mathbf{A} = \mathbf{C}(x^2+1) \oplus \mathbf{C}(x^2+1) \in M_4(\mathbb{Q})$ and $\mathbf{B} = \mathbf{C}((x^2+1)^2) \in M_4(\mathbb{Q})$ is semisimple? Is either diagonalizable over $\mathbb{Q}$?

Solution. $\mathbf{A}$ is semisimple. Each block has minimal polynomial x^2+1 (Theorem 10.5.3), so $m_{\mathbf{A}} = \operatorname{lcm}(x^2+1, x^2+1) = x^2+1$ (Proposition 9.5.6), an irreducible polynomial to the first power. $\mathbf{B}$ is not: $m_{\mathbf{B}} = (x^2+1)^2$ repeats the irreducible x^2+1. Neither is diagonalizable over $\mathbb{Q}$, since x^2+1 has no rational root, so neither minimal polynomial is a product of linear factors. Over $\mathbb{C}$, however, $\mathbf{A}$ becomes $\operatorname{diag}(i,-i,i,-i)$, while $\mathbf{B}$ becomes $\mathbf{J}_2(i) \oplus \mathbf{J}_2(-i)$, which is not diagonal.

10.8.2 Splitting an operator in two

Here is the goal. **When p_T splits, T is the sum of a diagonalizable operator and a nilpotent operator that commute, in exactly one way, and both summands are polynomials in T.** The Jordan form already shows *a* splitting: in a Jordan basis, $\mathbf{J} = \mathbf{D} + \mathbf{M}$ where $\mathbf{D}$ is the diagonal of $\mathbf{J}$ and $\mathbf{M}$ carries the superdiagonal ones, and these commute because $\mathbf{D}$ is constant on each block. What the theorem adds is that the two pieces do not depend on which Jordan basis was used, and that they can be written down without finding one.

Theorem 10.8.3. Jordan–Chevalley Decomposition

Let V be a finite-dimensional vector space over F with $\dim V = n \geq 1$, and let $T \in \mathcal{L}(V)$ with p_T split over F. Then:

(a) **(Existence)** There are $S, N \in \mathcal{L}(V)$ with

$$T = S + N, \qquad S \text{ diagonalizable}, \qquad N \text{ nilpotent}, \qquad SN = NS;$$

moreover $S = g(T)$ and $N = h(T)$ for some $g, h \in F[x]$.

(b) **(Uniqueness)** If $T = S' + N'$ with S' diagonalizable, N' nilpotent, and $S'N' = N'S'$, then $S' = S$ and $N' = N$.

We call S the **semisimple part** and N the **nilpotent part** of T.

Idea. **Existence.** The generalized eigenspace decomposition already puts the answer in front of us: on G_λ the operator is $\lambda\,\mathrm{id}$ +nilpotent. Declare S to be $\lambda\,\mathrm{id}$ on G_λ for each λ and let N be the rest. Then S is diagonalizable because a basis assembled from bases of the G_λ consists of eigenvectors of S, and N is nilpotent because it is nilpotent on each piece. That S and N are polynomials in T is the extra information carried by the primary decomposition: its projections are $h_i(T)$.

Uniqueness. A second decomposition $S' + N'$ has S' commuting with T, hence with every polynomial in T, hence with S and with N; the same for N'. Then $S - S' = N' - N$ is simultaneously diagonalizable and nilpotent, and the only such operator is 0.

Proof.

(a) Let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of T and write

$$m_T = (x-\lambda_1)^{s_1} \cdots (x-\lambda_k)^{s_k}, \qquad G_i = G_{\lambda_i}(T),$$

as in Theorem 10.1.3, so that $V = G_1 \oplus \cdots \oplus G_k$ with $G_i = \ker(T - \lambda_i \operatorname{id}_V)^{s_i}$. The factors $(x-\lambda_i)^{s_i}$ are exactly the primary factors of m_T, since $x - \lambda_i$ is irreducible, so G_i is the i-th primary component of Theorem 9.8.1. By part (d) of that theorem there are $h_1, \dots, h_k \in F[x]$ such that

$$\pi_i \coloneqq h_i(T) \in \mathcal{L}(V)$$

is the projection onto G_i along $\bigoplus_{j \neq i} G_j$. Put

$$g \coloneqq \sum_{i=1}^{k} \lambda_i h_i \in F[x],$$
$$S \coloneqq g(T) = \sum_{i=1}^{k} \lambda_i \pi_i,$$
$$N \coloneqq T - S = h(T) \text{ with } h = x - g.$$

S is diagonalizable. Fix i and $v \in G_i$. Then $\pi_i v = v$ and $\pi_j v = 0$ for $j \neq i$ (Theorem 4.5.1), so $Sv = \lambda_i v$. Choose a basis of each G_i and let $\mathcal{B}$ be the concatenation, a basis of V because $V = \bigoplus_i G_i$. Every vector of $\mathcal{B}$ is an eigenvector of S, so $[S]^{\mathcal{B}}_{\mathcal{B}}$ is diagonal and S is diagonalizable (Theorem 9.4.1).
N is nilpotent. Fix i and $v \in G_i$. Then $Nv = Tv - \lambda_i v = (T - \lambda_i \operatorname{id}_V)v$, which lies in G_i because G_i is T-invariant, so by induction $N^j v = (T - \lambda_i \operatorname{id}_V)^j v$ for every $j \geq 0$. With $s = \max_i s_i$ we get $N^s v = (T - \lambda_i \operatorname{id}_V)^s v = 0$, since $(T - \lambda_i \operatorname{id}_V)^{s_i} v = 0$ and $s \geq s_i$. As every $v \in V$ is a sum of such vectors, $N^s = 0$ and N is nilpotent (Definition 10.2.1).
They commute. Both $S = g(T)$ and $N = h(T)$ are polynomials in T, and any two polynomials in T commute (Theorem 6.7.1 (c)). This proves (a).

(b) Suppose $T = S' + N'$ as stated. First, S' commutes with T:

$$S'T = S'(S' + N') = S'S' + S'N' = S'S' + N'S' = (S' + N')S' = TS',$$

where the middle equality is the hypothesis $S'N' = N'S'$. Hence S' commutes with every power of T and so with every polynomial in T, in particular with $S = g(T)$ and $N = h(T)$. The same computation gives $N'T = TN'$, so N' too commutes with S and N.

Now set $X \coloneqq S - S'$. From $S + N = T = S' + N'$ we get

$$X = S - S' = N' - N.$$

The operators S and S' are diagonalizable and commute, so they are simultaneously diagonalizable (Theorem 9.9.5): there is a basis $\mathcal{C}$ in which $[S]^{\mathcal{C}}_{\mathcal{C}}$ and $[S']^{\mathcal{C}}_{\mathcal{C}}$ are both diagonal. Then $[X]^{\mathcal{C}}_{\mathcal{C}} = [S]^{\mathcal{C}}_{\mathcal{C}} - [S']^{\mathcal{C}}_{\mathcal{C}}$ is diagonal, so X is diagonalizable. On the other hand N and N' are nilpotent and commute, hence so do N' and $-N$. Say $N^a = 0$ and $(N')^b = 0$ with $a, b \geq 1$, and put $s = a + b - 1$. By distributivity, $X^s = (N' - N)^s$ is the sum of the 2^s products of s factors, each factor N' or $-N$; since the two commute, each product can be rearranged, one adjacent swap at a time, into $\pm (N')^i N^{\,s-i}$, where i counts the factors equal to N'. If $i \geq b$,

then $(N')^i = 0$; and if $i \le b-1$, then $s - i \ge s - b + 1 = a$, so $N^{s-i} = 0$. Either way the product is 0, so $X^s = 0$ and X is nilpotent.

So X is diagonalizable and nilpotent. Being nilpotent, $\operatorname{spec}(X) = \{0\}$ (Proposition 10.2.1 (c)); being diagonalizable, $[X]_{\mathcal{D}}^{\mathcal{D}}$ is diagonal with the eigenvalues of X on the diagonal for some basis $\mathcal{D}$ (Theorem 9.4.1), so $[X]_{\mathcal{D}}^{\mathcal{D}} = \mathbf{O}$ and $X = 0$. Hence $S' = S$ and, subtracting from T, $N' = N$. This proves the theorem.

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Warning. **Splitting a matrix into its diagonal part and its off-diagonal part is *not* the Jordan–Chevalley decomposition.** For $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$, the "obvious" split is $\mathbf{D} = \operatorname{diag}(1,2)$ plus $\mathbf{E} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Here $\mathbf{D}$ is diagonal and $\mathbf{E}$ is nilpotent, but

$$\mathbf{DE} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = \mathbf{ED},$$

so the commuting requirement fails, and the split is not the one the theorem produces. Indeed $\mathbf{A}$ has distinct eigenvalues, so it is diagonalizable (Corollary 9.4.2) and its actual decomposition is $\mathbf{S} = \mathbf{A}$, $\mathbf{N} = \mathbf{O}$. The off-diagonal part of a matrix is a feature of the basis; the nilpotent part is a feature of the operator.

The formulas in the proof are also the recipe: build the projections π_i onto the generalized eigenspaces — by Bézout, as in Example (A Primary Decomposition with an Irreducible Quadratic) — and set $S = \sum_i \lambda_i \pi_i$.

Example. The Semisimple and Nilpotent Parts of a 3×3 Matrix

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 2 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 2 \end{pmatrix} \in M_3(\mathbb{Q}), \qquad p_{\mathbf{A}} = (x-2)^2(x-3).$$

Find the Jordan–Chevalley decomposition $\mathbf{A} = \mathbf{S} + \mathbf{N}$, and exhibit $\mathbf{S}$ and $\mathbf{N}$ as polynomials in $\mathbf{A}$.

Solution. The minimal polynomial. Put $\mathbf{B} = \mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 0 & 2 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 0 \end{pmatrix}$. Then

$$\mathbf{B}(\mathbf{A} - 3\mathbf{I}) = \begin{pmatrix} 0 & 2 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 2 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 1 & -1 \\ 1 & -1 & 1 \\ 2 & -2 & 2 \end{pmatrix} \neq \mathbf{O},$$

so $m_{\mathbf{A}} \neq (x-2)(x-3)$; since $m_{\mathbf{A}} \mid p_{\mathbf{A}}$ and $m_{\mathbf{A}}$ has both 2 and 3 as roots (Theorem 9.5.3), $m_{\mathbf{A}} = (x-2)^2(x-3)$. The primary factors are $(x-2)^2$ and $x-3$.

Bézout. Dividing, $(x-2)^2 = x^2 - 4x + 4 = (x-3)(x-1) + 1$, so

$$1 \cdot (x-2)^2 - (x-1)(x-3) = 1.$$

Following Theorem 6.7.5 with $p = (x-2)^2$ and $q = x - 3$: the projection onto $\ker q(\mathbf{A}) = G_3(\mathbf{A})$ along $\ker p(\mathbf{A}) = G_2(\mathbf{A})$ is $\pi_3 = (\mathbf{A} - 2\mathbf{I})^2 = \mathbf{B}^2$, and $\pi_2 = \mathbf{I} - \mathbf{B}^2$.

Compute. Squaring,

$$\mathbf{B}^2 = \begin{pmatrix} -1 & 3 & -2 \\ 0 & 0 & 0 \\ 1 & -3 & 2 \end{pmatrix}, \qquad \text{and } \mathbf{B}^3 = \mathbf{B}^2 \text{ (as } \mathbf{B}^2(\mathbf{B} - \mathbf{I}) = \mathbf{O}).$$

Hence

$$\mathbf{S} = 2\pi_2 + 3\pi_3 = 2\mathbf{I} + \mathbf{B}^2 = \begin{pmatrix} 1 & 3 & -2 \\ 0 & 2 & 0 \\ 1 & -3 & 4 \end{pmatrix},$$

$$\mathbf{N} = \mathbf{A} - \mathbf{S} = \mathbf{B} - \mathbf{B}^2 = \begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ -2 & 2 & -2 \end{pmatrix}.$$

The checks. First $\mathbf{S} + \mathbf{N} = \mathbf{A}$ by construction. Next $\mathbf{N}^2 = \mathbf{O}$: reading the rows of $\mathbf{N}$ as multiples of $(1, -1, 1)$ shows that $\mathbf{N} = \mathbf{u}\mathbf{w}^\top$ with $\mathbf{u} = (1, -1, -2)$ and $\mathbf{w} = (1, -1, 1)$, and $\mathbf{w}^\top\mathbf{u} = 1 + 1 - 2 = 0$, so $\mathbf{N}^2 = \mathbf{u}(\mathbf{w}^\top\mathbf{u})\mathbf{w}^\top = \mathbf{O}$. Next $\mathbf{S}$ is diagonalizable: $(\mathbf{S} - 2\mathbf{I})(\mathbf{S} - 3\mathbf{I}) = \mathbf{B}^2(\mathbf{B}^2 - \mathbf{I}) = \mathbf{B}^4 - \mathbf{B}^2 = \mathbf{O}$, because $\mathbf{B}^3 = \mathbf{B}^2$ gives $\mathbf{B}^4 = \mathbf{B}^2$; so $m_{\mathbf{S}}$ divides a product of distinct linear factors and $\mathbf{S}$ is diagonalizable (Theorem 9.8.2). Finally $\mathbf{SN} = \mathbf{NS}$, since both are polynomials in $\mathbf{A}$: explicitly $\mathbf{S} = 2\mathbf{I} + (\mathbf{A} - 2\mathbf{I})^2$ and $\mathbf{N} = (\mathbf{A} - 2\mathbf{I}) - (\mathbf{A} - 2\mathbf{I})^2$, that is,

$$\mathbf{S} = g(\mathbf{A}) \text{ with } g = x^2 - 4x + 6,$$
$$\mathbf{N} = h(\mathbf{A}) \text{ with } h = -x^2 + 5x - 6 = -(x-2)(x-3).$$

As a cross-check, $g + h = x$, so $g(\mathbf{A}) + h(\mathbf{A}) = \mathbf{A}$; and $\operatorname{tr}\mathbf{S} = 1 + 2 + 4 = 7 = 2 + 2 + 3$ is the sum of the eigenvalues of $\mathbf{A}$, while $\operatorname{tr}\mathbf{N} = 1 + 1 - 2 = 0$, as it must be for a nilpotent matrix (Proposition 10.2.1 (c) and Theorem 9.3.3).

Two consequences are worth naming now. Because S and N are polynomials in T, **every operator commuting with T automatically commutes with S and with N.** And the decomposition is exactly what makes functions of matrices computable in the next section: any reasonable function f will satisfy $f(T) = f(S + N)$ with the two parts handled separately, S by its eigenvalues and N by finitely many terms, since its powers stop.

Remark. The splitting exists in greater generality. Over a perfect field, every T can be written uniquely as $T = S + N$ with S **semisimple** in the sense of Definition 10.8.1, N nilpotent, and $SN = NS$; the proof runs through the primary decomposition of m_T into powers of general irreducibles instead of linear ones. We prove only the split case, which is all we use.

10.8.3 Exercises

10.8.3.1 A. Check your understanding

Exercise. A1

(a) Define a semisimple operator, and say how the definition specializes to Chapter 9's criterion for diagonalizability.

(b) State the Jordan–Chevalley Decomposition Theorem, including all three conditions on S and N and the uniqueness clause.

(c) True or false: $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$ is semisimple. Justify your answer.

(d) True or false: if $\mathbf{A} = \mathbf{D} + \mathbf{E}$ with $\mathbf{D}$ the diagonal part of $\mathbf{A}$ and $\mathbf{E}$ the off-diagonal part, then $\mathbf{D}$ is the semisimple part of $\mathbf{A}$. Justify your answer.

(e) Which hypothesis of the uniqueness clause does the pair $(\mathbf{D}, \mathbf{E})$ of part (d) typically violate?

(f) Why does every operator commuting with T commute with the semisimple part of T?

Solution.

(a) T is semisimple if m_T is a product of **distinct monic irreducible** polynomials over F (Definition 10.8.1). Requiring those irreducibles to be **linear** gives Theorem 9.8.2, the criterion for diagonalizability. So diagonalizable $\Rightarrow$ semisimple, and the converse holds when m_T splits.

(b) If p_T splits, then $T = S + N$ with S diagonalizable, N nilpotent, and $SN = NS$; such S and N are unique, and both are polynomials in T (Theorem 10.8.3).

(c) True. Its minimal polynomial is $x^2 + 1$, which has no real root and so is irreducible over $\mathbb{R}$ (Theorem 6.3.1); one irreducible factor, exponent 1.

(d) False. For $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ the semisimple part is $\mathbf{A}$ itself, since $\mathbf{A}$ is diagonalizable, whereas $\mathbf{D} = \operatorname{diag}(1, 2) \neq \mathbf{A}$.

(e) The commuting condition $\mathbf{DE} = \mathbf{ED}$; see the warning after Theorem 10.8.3.

(f) Because $S = g(T)$ for some polynomial g (Theorem 10.8.3 (a)), and an operator commuting with T commutes with every power of T, hence with every polynomial in T.

10.8.3.2 B. Practice

Exercise. B1: Semisimple and nilpotent parts

For each matrix over $\mathbb{Q}$, find the Jordan–Chevalley decomposition $\mathbf{A} = \mathbf{S} + \mathbf{N}$ and express $\mathbf{S}$ as a polynomial in $\mathbf{A}$.

(a) $\mathbf{A}_1 = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$.

(b) $\mathbf{A}_2 = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$.

(c) $\mathbf{A}_3 = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$.

Solution.

(a) $m_{\mathbf{A}_1} = (x-3)^2$, so there is one eigenvalue and $G_3 = \mathbb{Q}^2$; the projection onto it is $\mathbf{I}$. Hence $\mathbf{S} = 3\mathbf{I}$ and $\mathbf{N} = \mathbf{A}_1 - 3\mathbf{I} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, with $\mathbf{N}^2 = \mathbf{O}$. As a polynomial, $\mathbf{S} = 3\mathbf{I} = g(\mathbf{A}_1)$ with $g = 3$ the constant polynomial. (Every scalar multiple of $\mathbf{I}$ is diagonalizable, and $\mathbf{S}$ commutes with everything.)

(b) $\mathbf{A}_2$ has the distinct eigenvalues 1 and 3, so it is diagonalizable (Corollary 9.4.2). By the uniqueness in Theorem 10.8.3, $\mathbf{S} = \mathbf{A}_2$ and $\mathbf{N} = \mathbf{O}$; here $g = x$. The naive split into $\operatorname{diag}(1,3)$ and $\begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$ is **not** the Jordan–Chevalley decomposition.

(c) $p_{\mathbf{A}_3} = (x-2)^2(x-5)$. Put $\mathbf{M} = \mathbf{A}_3 - 2\mathbf{I} = \begin{pmatrix} 0 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}$. Then $\mathbf{M}(\mathbf{A}_3 - 5\mathbf{I})$ has $(1,2)$-entry $-3 \neq 0$, so $m_{\mathbf{A}_3} = (x-2)^2(x-5)$. Bézout: $(x-2)^2 = (x-5)(x+1)+9$, so $\frac{1}{9}(x-2)^2 - \frac{1}{9}(x+1)(x-5) = 1$, and by Theorem 6.7.5 the projection onto G_5 along G_2 is $\pi_5 = \frac{1}{9}\mathbf{M}^2$. Computing $\mathbf{M}^2 = \begin{pmatrix} 0 & 0 & 9 \\ 0 & 0 & 0 \\ 0 & 0 & 9 \end{pmatrix}$,

$$\mathbf{S} = 2\mathbf{I} + 3\pi_5 = 2\mathbf{I} + \tfrac{1}{3}\mathbf{M}^2 = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}, \qquad \mathbf{N} = \mathbf{A}_3 - \mathbf{S} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Checks: $\mathbf{N}^2 = \mathbf{O}$; $\mathbf{S} - 2\mathbf{I}$ has rank 1, so $g_{\mathbf{S}}(2) = 2 = a_{\mathbf{S}}(2)$ and $\mathbf{S}$ is diagonalizable (Theorem 9.4.1); and $\mathbf{SN} = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \mathbf{NS}$. As a polynomial, $\mathbf{S} = 2\mathbf{I} + \frac{1}{3}(\mathbf{A}_3 - 2\mathbf{I})^2 = g(\mathbf{A}_3)$ with $g = \frac{1}{3}x^2 - \frac{4}{3}x + \frac{10}{3}$. Note that $\mathbf{S}$ keeps the entry 3 in position $(1,3)$: the diagonal part of $\mathbf{A}_3$ is the wrong answer, and it does not commute with the rest, since $\operatorname{diag}(2,2,5)(\mathbf{A}_3 - \operatorname{diag}(2,2,5))$ and the reverse product differ in position $(1,3)$ by -9.

Exercise. B2: When is the nilpotent part zero?

Let V be finite-dimensional with $\dim V \geq 1$ and let $T \in \mathcal{L}(V)$ with p_T split, with Jordan–Chevalley decomposition $T = S + N$. Prove that $N = 0$ if and only if T is diagonalizable, and that $S = 0$ if and only if T is nilpotent.

Solution. If $N = 0$, then $T = S$ is diagonalizable. Conversely, if T is diagonalizable, then $T = T + 0$ is a decomposition into a diagonalizable and a nilpotent part that commute ($T \cdot 0 = 0 = 0 \cdot T$), so by the uniqueness in Theorem 10.8.3, $S = T$ and $N = 0$.

If $S = 0$, then $T = N$ is nilpotent. Conversely, if T is nilpotent, then $T = 0 + T$ is such a decomposition, since the zero operator is diagonalizable and commutes with T; uniqueness gives $S = 0$ and $N = T$.

Exercise. B3: Block diagonal matrices

Let $\mathbf{A}_1 \in M_p(F)$ and $\mathbf{A}_2 \in M_q(F)$ have split characteristic polynomials, with Jordan–Chevalley decompositions $\mathbf{A}_i = \mathbf{S}_i + \mathbf{N}_i$. Prove that the Jordan–Chevalley decomposition of $\mathbf{A}_1 \oplus \mathbf{A}_2$ is

$$\mathbf{A}_1 \oplus \mathbf{A}_2 = (\mathbf{S}_1 \oplus \mathbf{S}_2) + (\mathbf{N}_1 \oplus \mathbf{N}_2).$$

Solution. Write $\mathbf{S} = \mathbf{S}_1 \oplus \mathbf{S}_2$ and $\mathbf{N} = \mathbf{N}_1 \oplus \mathbf{N}_2$. By Theorem 8.1.4 (a), $\mathbf{S} + \mathbf{N} = (\mathbf{S}_1 + \mathbf{N}_1) \oplus (\mathbf{S}_2 + \mathbf{N}_2) = \mathbf{A}_1 \oplus \mathbf{A}_2$. Also $p_{\mathbf{A}_1 \oplus \mathbf{A}_2} = p_{\mathbf{A}_1} p_{\mathbf{A}_2}$ splits (Theorem 7.8.4, applied over $F[x]$ as the remark after it allows), so Theorem 10.8.3 applies to $\mathbf{A}_1 \oplus \mathbf{A}_2$ and it suffices, by uniqueness, to check the three conditions.

$\mathbf{S}$ *is diagonalizable.* Pick invertible $\mathbf{P}_i$ with $\mathbf{P}_i^{-1}\mathbf{S}_i\mathbf{P}_i$ diagonal. Then $\mathbf{P} = \mathbf{P}_1 \oplus \mathbf{P}_2$ is invertible with $\mathbf{P}^{-1} = \mathbf{P}_1^{-1} \oplus \mathbf{P}_2^{-1}$, and $\mathbf{P}^{-1}\mathbf{S}\mathbf{P} = (\mathbf{P}_1^{-1}\mathbf{S}_1\mathbf{P}_1) \oplus (\mathbf{P}_2^{-1}\mathbf{S}_2\mathbf{P}_2)$ is diagonal (Theorem 8.1.4 (a), (c)).

$\mathbf{N}$ *is nilpotent.* If $\mathbf{N}_1^a = \mathbf{O}$ and $\mathbf{N}_2^b = \mathbf{O}$, then $\mathbf{N}^{\max(a,b)} = \mathbf{N}_1^{\max(a,b)} \oplus \mathbf{N}_2^{\max(a,b)} = \mathbf{O}$ by Theorem 8.1.4 (b).

They commute. $\mathbf{S}\mathbf{N} = (\mathbf{S}_1\mathbf{N}_1) \oplus (\mathbf{S}_2\mathbf{N}_2) = (\mathbf{N}_1\mathbf{S}_1) \oplus (\mathbf{N}_2\mathbf{S}_2) = \mathbf{N}\mathbf{S}$, again by Theorem 8.1.4 (a).

By the uniqueness in Theorem 10.8.3, $\mathbf{S}$ and $\mathbf{N}$ are the semisimple and nilpotent parts of $\mathbf{A}_1 \oplus \mathbf{A}_2$.

10.8.3.3 C. Going deeper

Exercise. C1: The multiplicative version

Call $U \in \mathcal{L}(V)$ **unipotent** if $U - \mathrm{id}_V$ is nilpotent. Let $T \in \mathcal{L}(V)$ be **invertible** with p_T split, and let $T = S + N$ be its Jordan–Chevalley decomposition.

(a) Prove that S is invertible.

(b) Prove that S^{-1} commutes with N, and that $S^{-1}N$ is nilpotent.

(c) Deduce that $T = SU$ with S diagonalizable, U unipotent, and $SU = US$.

(d) Prove that this multiplicative decomposition is unique: if $T = S'U'$ with S' diagonalizable, U' unipotent, and $S'U' = U'S'$, then $S' = S$ and $U' = U$.

Hint: for (d), turn the multiplicative statement into an additive one by writing $T = S' + S'(U' - \mathrm{id}_V)$.

Solution.

(a) By the proof of Theorem 10.8.3, $Sv = \lambda_i v$ for every $v \in G_{\lambda_i}(T)$, so $\mathrm{spec}(S) = \mathrm{spec}(T)$ — a basis of eigenvectors of S with these eigenvalues was exhibited there. Since T is

invertible, $0 \notin \operatorname{spec}(T)$ (Theorem 9.2.6), so $0 \notin \operatorname{spec}(S)$ and S is invertible.

(b) From $SN = NS$, multiply on both sides by S^{-1}: $S^{-1}(SN)S^{-1} = S^{-1}(NS)S^{-1}$ gives $NS^{-1} = S^{-1}N$. Hence $(S^{-1}N)^j = S^{-j}N^j$ for every $j \geq 1$, by induction using $NS^{-1} = S^{-1}N$. With $N^s = 0$ we get $(S^{-1}N)^s = 0$.

(c) Put $U = \mathrm{id}_V + S^{-1}N$. Then $SU = S + N = T$, and $U - \mathrm{id}_V = S^{-1}N$ is nilpotent by (b), so U is unipotent. They commute: $SU = S + N = US$, since $S(S^{-1}N) = N = (S^{-1}N)S$ by (b).

(d) Suppose $T = S'U'$ with S' diagonalizable, U' unipotent, and $S'U' = U'S'$. Put $N' = S'(U' - \mathrm{id}_V)$, so that $T = S' + N'$. Then N' is nilpotent: S' commutes with U', hence with $U' - \mathrm{id}_V$, so $(N')^j = (S')^j(U' - \mathrm{id}_V)^j = 0$ once j is large. And $S'N' = S'S'(U' - \mathrm{id}_V) = S'(U' - \mathrm{id}_V)S' = N'S'$. So $T = S' + N'$ satisfies the hypotheses of Theorem 10.8.3 (b), giving $S' = S$ and $N' = N$. Then $U' - \mathrm{id}_V = (S')^{-1}N' = S^{-1}N = U - \mathrm{id}_V$, so $U' = U$.

Exercise. C2: The parts inherit every symmetry

Let V be finite-dimensional with $\dim V \geq 1$, let $T \in \mathcal{L}(V)$ with p_T split, and let $T = S + N$ be its Jordan–Chevalley decomposition.

(a) Prove that if $X \in \mathcal{L}(V)$ satisfies $XT = TX$, then $XS = SX$ and $XN = NX$.

(b) Deduce that every T-invariant subspace is S-invariant and N-invariant, and that S and N map each $G_\lambda(T)$ into itself.

(c) Give an example of a diagonalizable S and a nilpotent N with $SN \neq NS$ for which $S + N$ is **not** the operator whose semisimple part is S. (So the commuting hypothesis cannot be dropped from the uniqueness statement.)

Solution.

(a) By Theorem 10.8.3 (a), $S = g(T)$ and $N = h(T)$ for some $g, h \in F[x]$. From $XT = TX$ we get $XT^j = T^jX$ for all $j \geq 0$ by induction, hence $Xf(T) = f(T)X$ for every $f \in F[x]$ by taking linear combinations. Applying this to $f = g$ and $f = h$ gives $XS = SX$ and $XN = NX$.

(b) Let $U \subseteq V$ be T-invariant. Then U is T^j-invariant for every j, hence $f(T)$-invariant for every $f \in F[x]$ (a sum of vectors of U lies in U); taking $f = g, h$ gives that U is S- and N-invariant. In particular $G_\lambda(T) = \ker(T - \lambda\,\mathrm{id}_V)^n$ is T-invariant (Corollary 10.1.2 (b)), so S and N map it into itself.

(c) Take $\mathbf{S} = \operatorname{diag}(1, 2)$ and $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ in $M_2(\mathbb{Q})$. Then $\mathbf{S}$ is diagonal, $\mathbf{N}^2 = \mathbf{O}$, and $\mathbf{SN} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = \mathbf{NS}$. Their sum $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ has distinct eigenvalues, so it is diagonalizable, and its semisimple part is $\mathbf{A}$ itself, not $\mathbf{S}$, with nilpotent part $\mathbf{O}$, not $\mathbf{N}$ (Exercise (B2: When is the nilpotent part zero?)). Without the commuting hypothesis, an operator can be written as "diagonalizable plus nilpotent" in many ways: for any $c \in \mathbb{Q}$, $\mathbf{A} = \begin{pmatrix} 1 & c \\ 0 & 2 \end{pmatrix} + \begin{pmatrix} 0 & 1 - c \\ 0 & 0 \end{pmatrix}$ is such a splitting, the first summand being diagonalizable because its eigenvalues $1, 2$ are distinct.